Mumbai University Bachelor's of Computer Engineering (syllabus)

Semester VII

Course Code	Course Name	Module Name
CSC701	Machine Learning	1. Introduction to Machine Learning 2. Learning with Regression and Trees 3. Ensemble Learning 4. Learning with Classification 5. Learning with Clustering 6. Dimensionality Reduction
CSC702	Big Data Analytics	1. Introduction to Big Data and Hadoop 2. Hadoop HDFS and MapReduce 3. NoSQL 4. Mining Data Streams 5. Real-Time Big Data Models 6. Data Analytics with R
CSDC 701X	Department Level Optional Course-3	
CSDC7011	Machine Vision	1. Introduction to Machine Vision 2. Digital Image Fundamentals 3. Machine Vision and System Components 4. Digital Image Processing for Machine Vision Applications 5. Motion Analysis 6. Emerging Trends in Machine Vision
CSDC7012	Quantum Computing Processing	1. Introduction to Quantum Computing 2. Mathematical Foundations for Quantum Computing 3. Building Blocks for Quantum Program 4. Quantum Algorithms and Error correction 5. Quantum Hardware 6. OSS Toolkits for implementing Quantum program
CSDC7013	Natural Language Processing	1. Introduction to NLP 2. Word Level Analysis 3. Syntax analysis 4. Semantic Analysis 5. Pragmatic & Discourse Processing 6. Applications of NLP
CSDC 702X	Department Level Optional Course-4	

CSDC7021	Augmented and Virtual Reality	1. Introduction to Virtual Reality 2. The Geometry of Virtual Worlds 3. Motion in Real and Virtual Worlds 4. Applying Virtual Reality 5. Augmented Reality 6. AR Development & Applications
CSDC7022	Block Chain	1. Introduction to Blockchain 2. Cryptocurrency 3. Programming for Blockchain 4. Public Blockchain 5. Private Blockchain 6. Tools and Applications of Blockchain
CSDC7023	Information Retrieval	1. Introduction to Information Retrieval 2. Modeling in Information Retrieval 3. Query and Operations in Information Retrieval 4. Indexing and Scoring in Information Systems 5. Evaluation of Information Retrieval Systems 6. Applications of Information Retrieval Systems
ILO 701X	Institute Level Optional Course-1	
ILO7011	Product Lifecycle Management	1. Introduction to Product Lifecycle Management (PLM) 2. Product Design 3. Product Data Management (PDM) 4. Virtual Product Development Tools 5. Integration of Environmental Aspects in Product Design 6. Life Cycle Assessment and Life Cycle Cost Analysis
ILO7012	Reliability Engineering	1. Probability theory, Probability Distributions, Measures of Dispersion 2. Reliability Concepts, Failure Data Analysis, Reliability Hazard Models 3. System Reliability 4. Reliability Improvement 5. Maintainability and Availability 6. Failure Mode, Effects and Criticality Analysis
ILO7013	Management Information System	1. Introduction To Information Systems (IS) 2. Introduction To Information Systems (IS), Business intelligence (BI)

		3. Ethical issues and Privacy 4. Social Computing (SC) 5. Computer Networks Wired and Wireless technology, Pervasive computing, Cloud computing model. 6. Information System within Organization
ILO7014	Design of Experiments	1. Introduction 2. Fitting Regression Models 3. Two-Level Factorial Designs 4. Two-Level Fractional Factorial Designs 5. Response Surface Methods and Designs 6. Taguchi Approach
ILO7015	Operation Research	1. Introduction to Operations Research, Linear Programming, Transportation Problem, Assignment Problem, Integer Programming Problem 2. Queuing models 3. Simulation 4. Dynamic programming 5. Game Theory 6. Inventory Models
ILO7016	Cyber Security and Laws	1. Introduction to Cybercrime 2. Cyber offenses & Cybercrime 3. Tools and Methods Used in Cyberline 4. The Concept of Cyberspace 5. Indian IT Act. 6. Information Security Standard compliances
ILO7018	. Energy Audit and Management	1. Energy Scenario 2. Energy Audit Principles 3. Energy Management and Energy Conservation in Electrical System and Energy efficiency measures in lighting system, Lighting control 4. Energy Management and Energy Conservation in Thermal Systems 5. Energy Performance Assessment 6. Energy conservation in Buildings
ILO7019	Development Engineering	1. Introduction to Rural Development Meaning, nature and scope of development

		2. Post-Independence rural Development Balwant Rai Mehta Committee - three tier system of rural local Government 3. Rural Development Initiatives in Five Year Plans Five Year Plans and Rural Development 4. Post 73rd Amendment Scenario 73rd Constitution Amendment Act, including - XI schedule, devolution of powers, functions and finance 5. Values and Science and Technology Material development and its values 6. Ethics Canons of ethics; ethics of virtue; ethics of duty; ethics of responsibility; Work ethics; Professional ethics;

Semester VIII

Course code	Course Name	Module Name
CSC801	Distributed Computing	1. Introduction to Distributed Systems 2. Communication 3. Synchronization 4. Resource and Process Management 5. Replication, Consistency and Fault Tolerance 6. Distributed File Systems
CSDC 801X	Department Level Optional Course -5	
CSDC8011	CSDC8011 : Deep Learning	1. Fundamentals of Neural Network 2. Training, Optimization and Regularization of Deep Neural Network 3. Autoencoders: Unsupervised Learning 4. Convolutional Neural Networks (CNN): Supervised Learning 5. Recurrent Neural Networks (RNN) 6. Recent Trends and Applications
CSDC8012	CSDC8012 : Digital Forensic	1. Introduction to Digital Forensics 2. Digital Evidence, Forensics Duplication and Digital Evidence Acquisition 3. Forensics Investigation

		4. Windows and Unix Forensics Investigation 5. Mobile Forensics 6. Browser, Email Forensic & Forensic Investigation Reporting
CSDC8013	CSDC8013 : Applied Data Science	1. Introduction to Data Science 2. Data Exploration 3. Methodology and Data Visualization 4. Anomaly Detection 5. Time Series Forecasting 6. Applications of Data Science
CSDC 802X	Department Level Optional Course -6	
CSDC8021	Optimization in Machine Learning	1. Introduction and Background to Optimization Theory 2. Derivative based Optimization 3. Stochastic Methods 4. Convex Optimization 5. Evolutionary Methods 6. Advance Evolutionary Methods
CSDC8022	High Performance Computing	1. Introduction to Parallel Computing 2. Parallel Programming Platform and Algorithm Design 3. Performance Measures 4. HPC Programming: OpenMP and MPI 5. Parallel programming using accelerators 6. High Performance Computing in the Cloud
CSDC8023	Social Media Analytics	1. Social Media Analytics: An Overview 2. Social Network Structure, Measures & Visualization 3. Social Media Text, Action & Hyperlink Analytics 4. Social Media Location & Search Engine Analytics 5. Social Information Filtering 6. Social Media Analytics Applications and Privacy
ILO 801X	Institute Level Optional Course -2	
ILO8021	Project Management	1. Project Management Foundation 2. Initiating Projects 3. Project Planning and Scheduling 4. Planning Projects 5. Executing Projects, Monitoring and Controlling Projects, Project Contracting

		6. Project Leadership and Ethics, Closing the Project
ILO8022	.Finance Management	1. Overview of Indian Financial System, Financial Instruments, Financial Markets, Financial Institutions 2. Concepts of Returns and Risks, Time Value of Money 3. Overview of Corporate Finance, Financial Ratio Analysis 4. Capital Budgeting, Working Capital Management 5. Sources of Finance, Capital Structure 6. Dividend Policy
ILO8023.	Entrepreneurship Development and Management	1. Overview Of Entrepreneurship 2. Business Plans And Importance Of Capital To Entrepreneurship, Entrepreneurship And Business Development 3. Women's Entrepreneurship Development, Social entrepreneurship-role and need, EDP cell, role of sustainability and sustainable development for SMEs, case studies, exercises 4. Indian Environment for Entrepreneurship 5. Effective Management of Business 6. Achieving Success In The Small Business
ILO8024	. Human Resource Managemen	1. Introduction to HR 2. Organizational Behaviour (OB) 3. Organizational Structure &Design 4. Human resource Planning 5. Emerging Trends in HR 6. HR & MIS, Strategic HRM, Labor Laws & Industrial Relations
ILO8025.	Professional Ethics and CSR	1. Professional Ethics and Business 2. Professional Ethics in the Marketplace, Professional Ethics and the Environment 3. Professional Ethics of Consumer Protection, Professional Ethics of Job Discrimination 4. Introduction to Corporate Social Responsibility 5. Corporate Social Responsibility 6. Corporate Social Responsibility in Globalizing India
ILO8026.	Research Methodology	1. Introduction and Basic Research Concepts 2. Types of Research

		3. Research Design and Sample Design 4. Research Methodology 5. Formulating Research Problem 6. Outcome of Research
ILO8027	IPR and Patenting	1. Introduction to Intellectual Property Rights (IPR), Importance of IPR in Modern Global Economic Environment 2. Enforcement of Intellectual Property Rights, Indian Scenario of IPR 3. Emerging Issues in IPR 4. Basics of Patents 5. Patent Rules 6. Procedure for Filing a Patent (National and International), Patent databases
ILO8028.	Digital Business Management	1. Introduction to Digital Business, Drivers of digital business 2. Overview of E-Commerce 3. Digital Business Support services, Application Development 4. Managing E-Business 5. E-Business Strategy 6. Materializing e-business: From Idea to Realization
ILO8029.	Environmental Management	1. Introduction and Definition of Environment 2. Global Environmental concerns 3. Concepts of Ecology 4. Scope of Environment Management, Role and functions of Government as a planning and regulating agency 5. Total Quality Environmental Management, ISO-14000, EMS certification. 6. General overview of major legislations like Environment Protection Act